

# COLLABORATOR SIDE



Collaborator  $X_j$  + LiDAR

PointPillars / BEV backbone

shared weights

Object-level message  $L$

Feature codec

Feature-level message  $F$  (16-QAM)

$\approx 0.024$  Msym/frame

$\approx 0.99$  Msym/frame

C256: 256-QAM comparator (not deployed),  $\approx 0.495$  Msym/frame

dominated; excluded from deployment (§III-B)

select one, controlled by  $s$

(one message)

2-bit request (via CAM)

# EGO SIDE — ON-VEHICLE DECISION



Ego vehicle  $X_e$

“Selector runs on the ego vehicle”

LiDAR

PointPillars / BEV backbone

shared weights

Cue Extractor

21-d

Channel estimator

(SNR, dB)

(AWGN/ Rayleigh)

RF Selector

$g(z_t, \hat{y}_t, \hat{\sigma}_t) \mapsto S_t$

Fusion  $\Psi$

$F_{ego,t} + m_{s_t}$

Detection head  $D$

(3D boxes)

V2V wireless channel (SNR, BLER)

local detection output

$F_{ego,t}$

request: ego  $\rightarrow$  collaborator message: collaborator  $\rightarrow$  ego